International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 3

Roadmap

- Current account
  - Net international investment position
    - The valuation effect
    - Exorbitant tax priviledge
- BOP examples

CA and Net International Investment Position

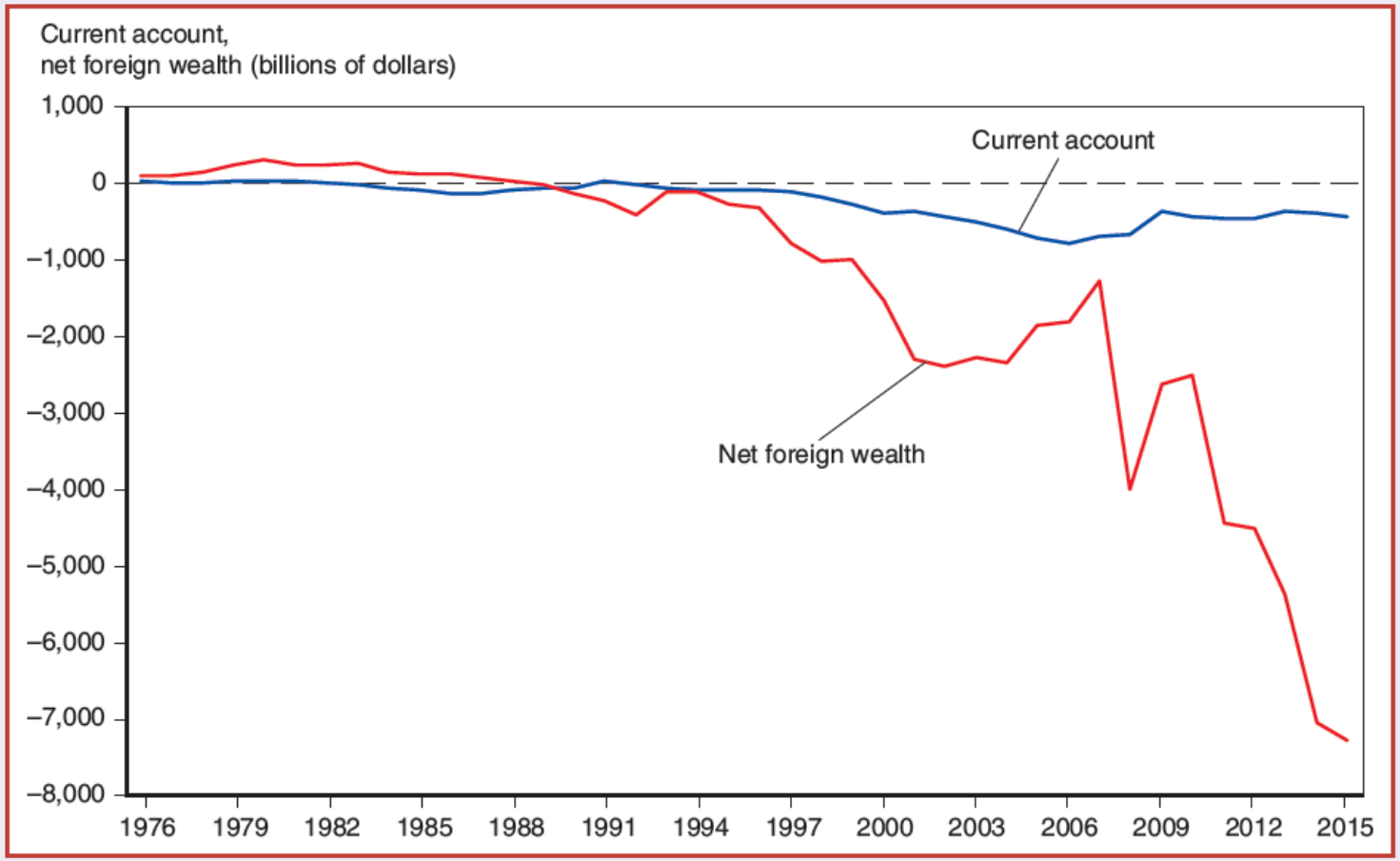
- *Net* international investment position (NIIP) is the difference between the value of foreign assets held by domestic residents and domestic assets held by foreign residents (i.e., domestic liabilities)

- Current account results in a change in NIIP

$$\Delta NIIP_t = NIIP_t - NIIP_{t-1} \approx CA_t$$

  - $NIIP_t$- the country's *net* international investment position at the end of period t
  - An approximation is due to *the valuation effect* (coming soon)

CA and Net International Investment Position



Pearson MyLab Economics Real-time data

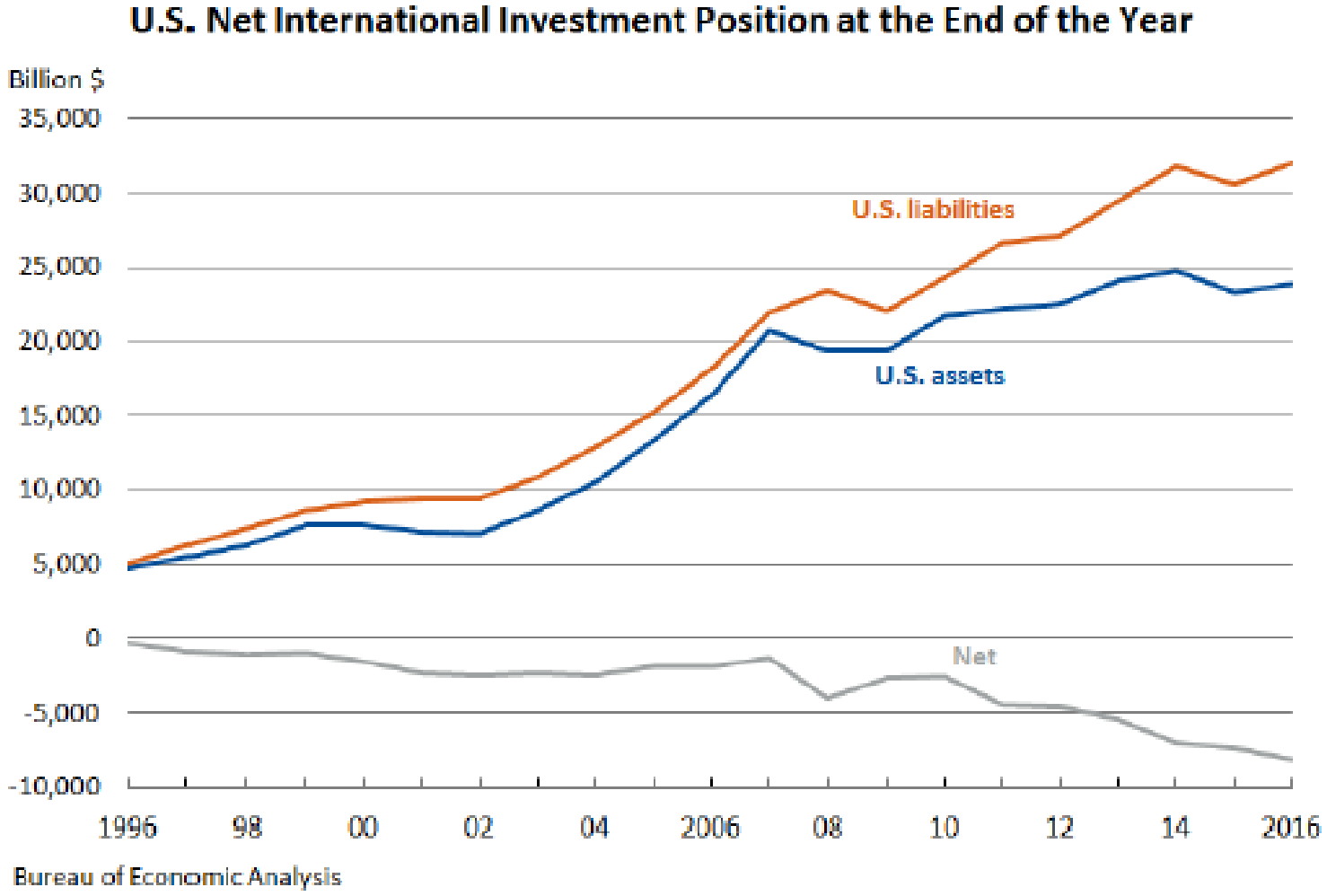
FIGURE 13-2

The U.S. Current Account and Net International Investment Position, 1976–2015

A string of current account deficits starting in the early 1980s reduced America's net foreign wealth until, by the early 21st century, the country had accumulated a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis.

CA and Net International Investment Position

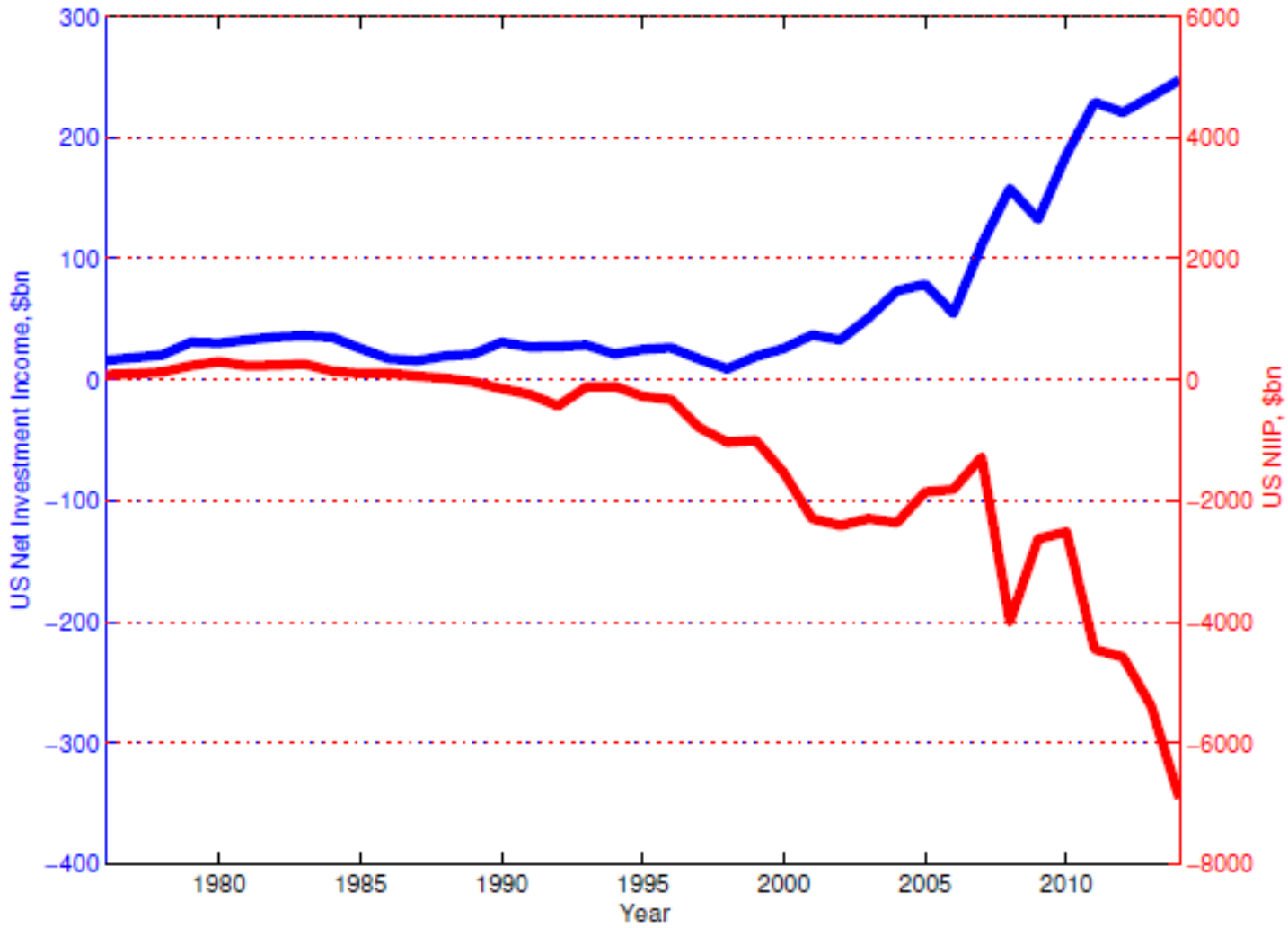


- Note the difference between *net* and *gross* **international investment position**

Question

If a country has a negative net international investment position, does it receive negative or positive return on this position?

Figure 1.7: Net Investment Income and the Net International Investment Position (United States 1976-2014)



Data Source: http://www.bea.gov

The Negative NIIP And Positive NII *Paradox*

- Net international investment position (NIIP)

$$NIIP = A - L$$

  – where A is country's foreign assets and L is country's foreign liabilities

- Net investment income (NII)

$$NII = r^A A - r^L L$$

  – where r^A is the interest on foreign assets, r^L is the interest on foreign liabilities

Gross International Investment Positions, U.S.

Right after WWII

1. NIIP/GDP=12%
2. Assets and liabilities are small
3. Assets>>Liabilities

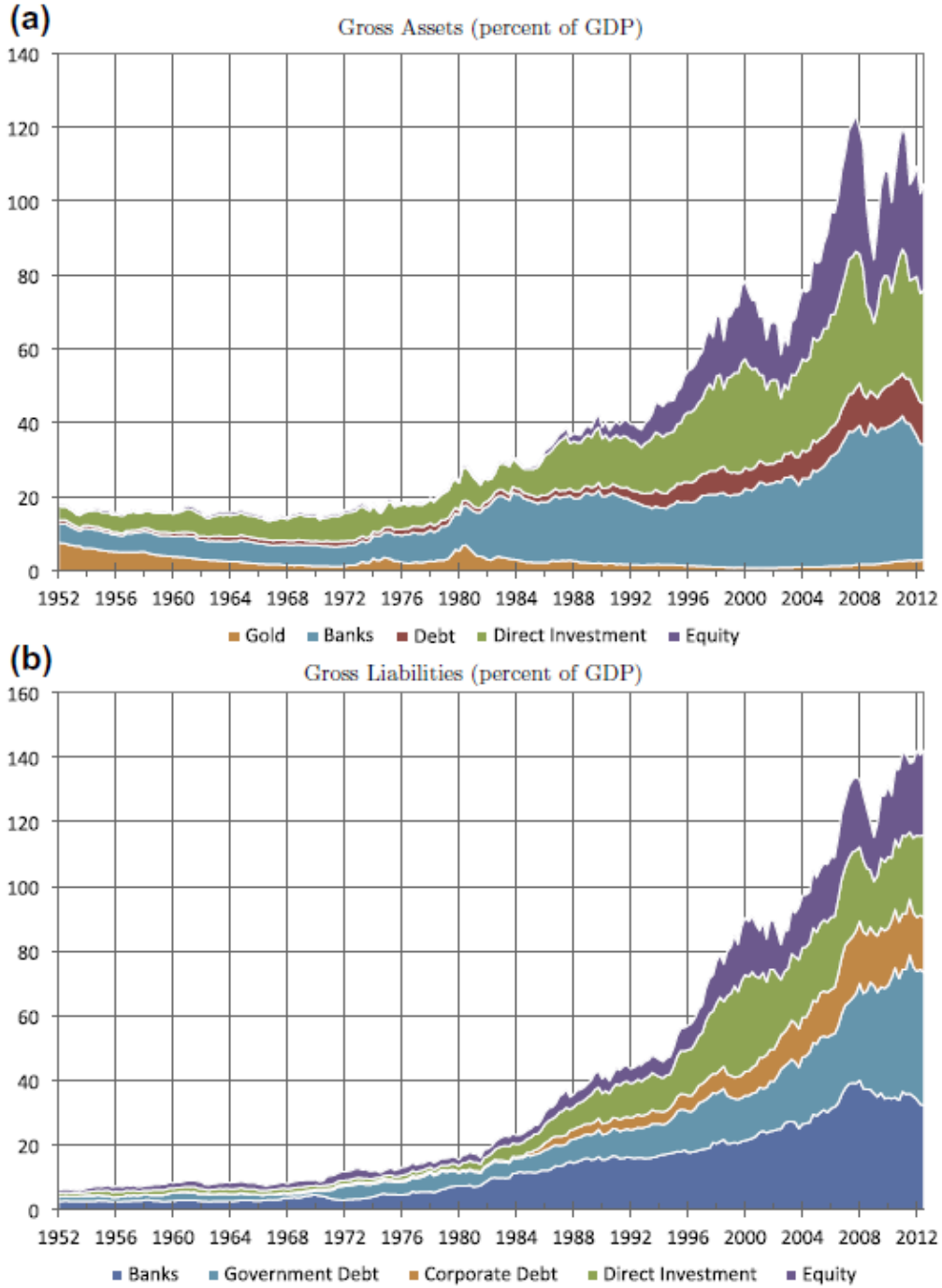


Figure 10.9 U.S. Gross Asset and Liabilities, by Asset Class, 1952–2012. *Source:* Gourinchas et al. (2010)

The 21st century

1. NIIP/GDP=-20% in 2010
2. Assets and liabilities are much bigger

Source: Gourinchas, Rey (2014) "External Adjustment, Global Imbalances, Valuation Effects" Section 5.1.3

US is like "a very leveraged investor"

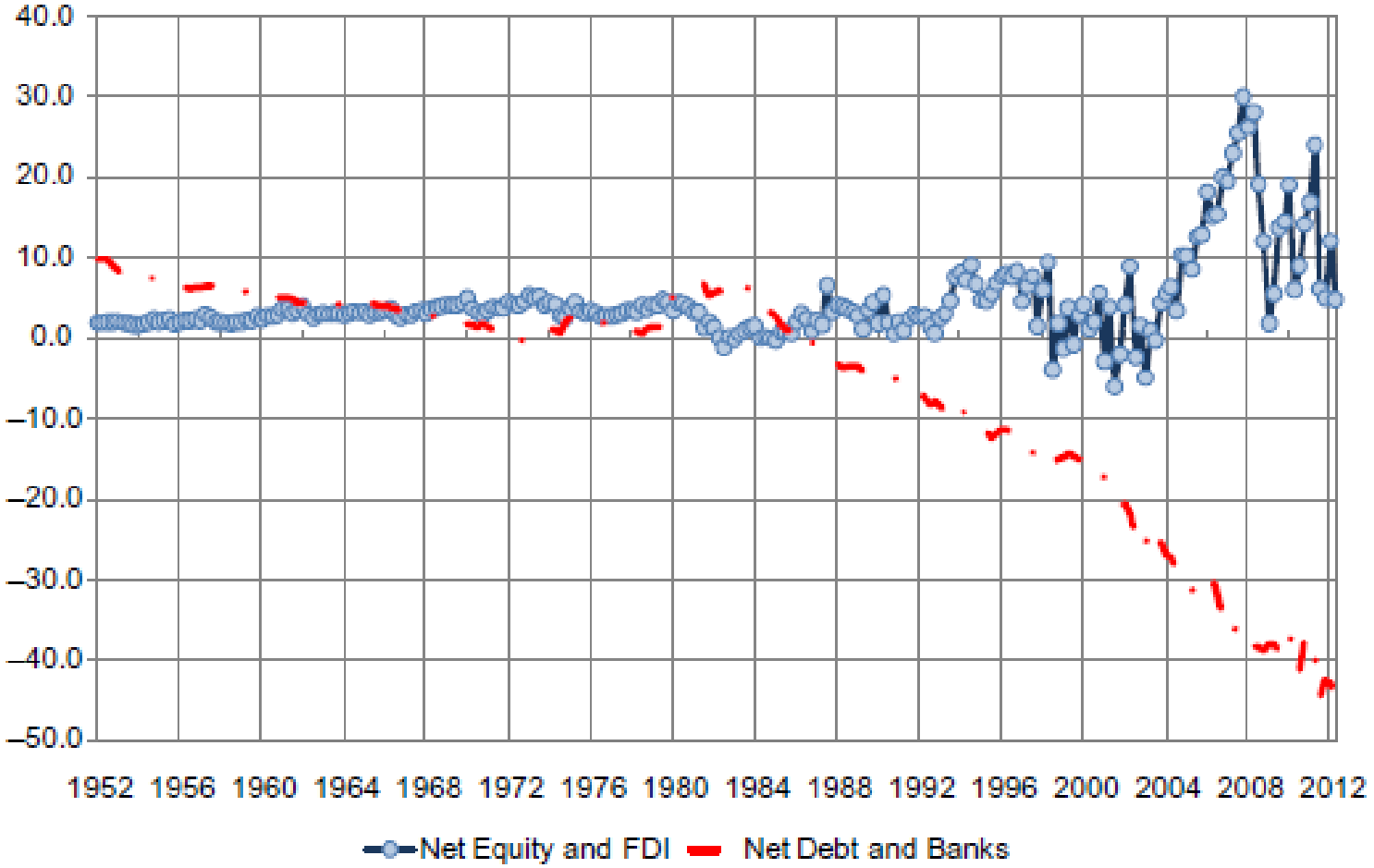
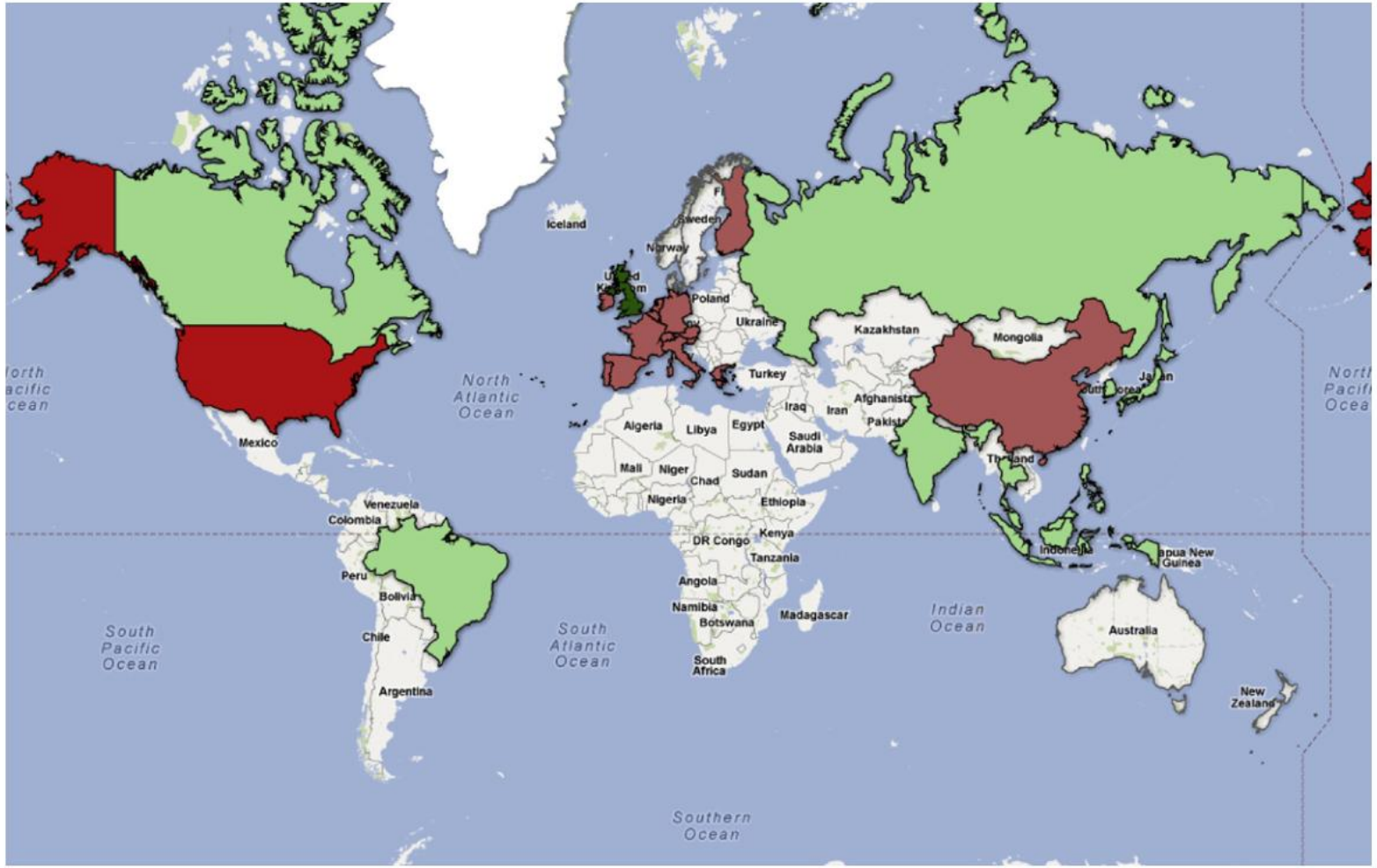


Figure 10.10 U.S. Net Portfolio Equity and Direct Investment (Percent of GDP) and Net Portfolio Debt and Other Assets (Percent GDP). *Source:* Gourinchas et al. (2010) Updated to 2012

- US is long on risky (*high* return) foreign assets and short on safe (*low* return) domestic assets
- The yield that the US received on its assets is higher than the yield that it paid on its liabilities by about 2% over 1952-2011. (This observation is often called ***exorbitant privilege***)

Wealth Transfers in 2008 (Exorbitant Duty)



Exorbitant duty – during global recessions US (the world's leveraged investor) suffers big losses due to

1. A decline in risky assets' value
2. Appreciation of the dollar (because the $ safe assets demand increases)

Fig. 1. Heat map of valuation gains and losses. The figure reports total valuation gains/losses. ■ Losses in excess of $400 billion. ■ Losses between $10 billion and $400 billion. ■ Gains between $10 billion and $400 billion. ■ Gains in excess of $400 billion.

Source: Gourinchas, Rey, Truempler (2012)

The Valuation Effect

$\Delta NIIP \approx CA$

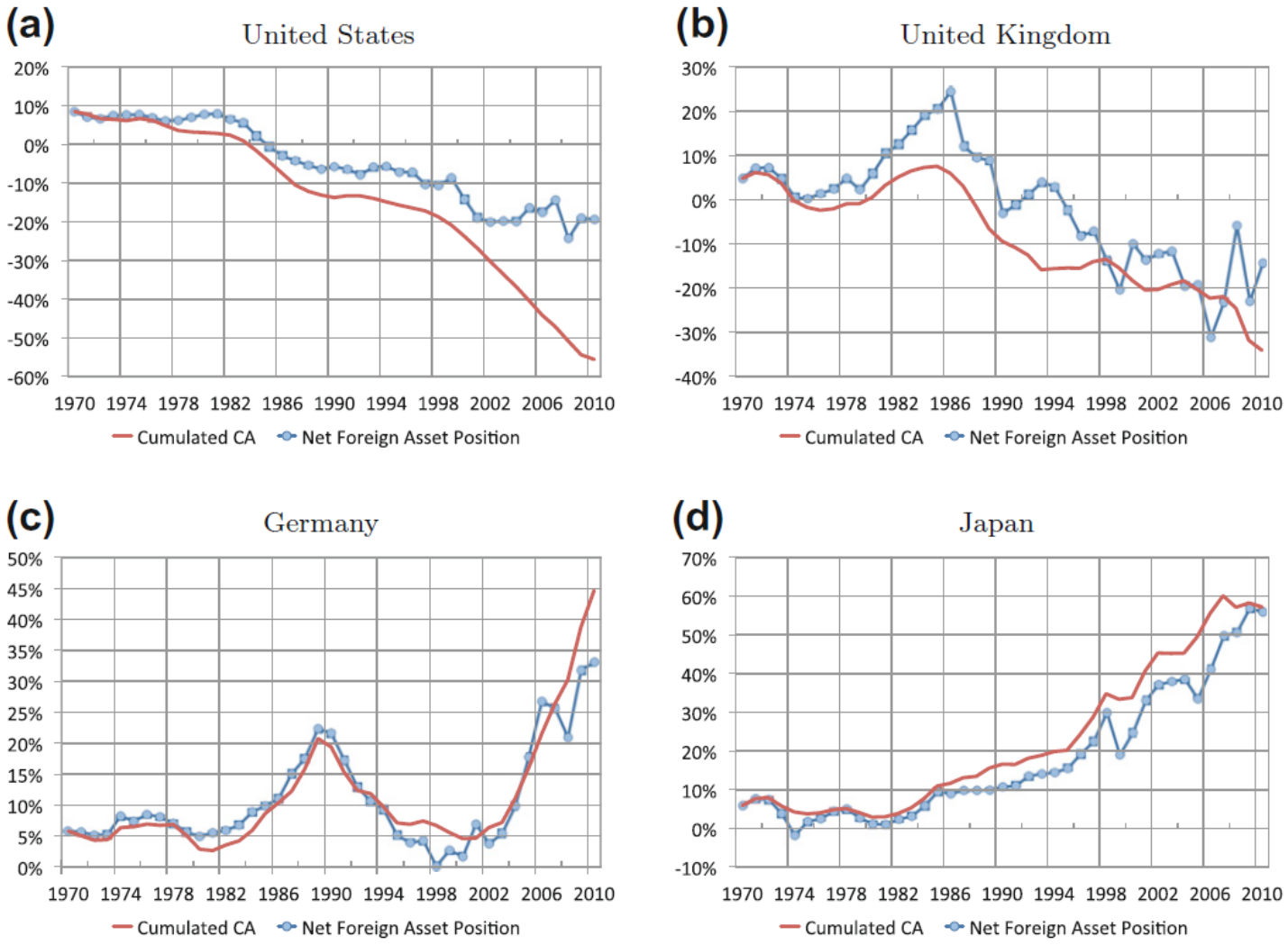


Figure 10.6 Cumulated Current Account and Net Foreign Asset Position: U.S., U.K., Germany, and Japan, 1970–2010. Percent of GDP. *Source:* Lane and Milesi-Ferretti (2007a) updated to 2010.

Source: Gourinchas, Rey (2014) "External Adjustment, Global Imbalances, Valuation Effects" Section 2

$\Delta NIIP \approx CA$

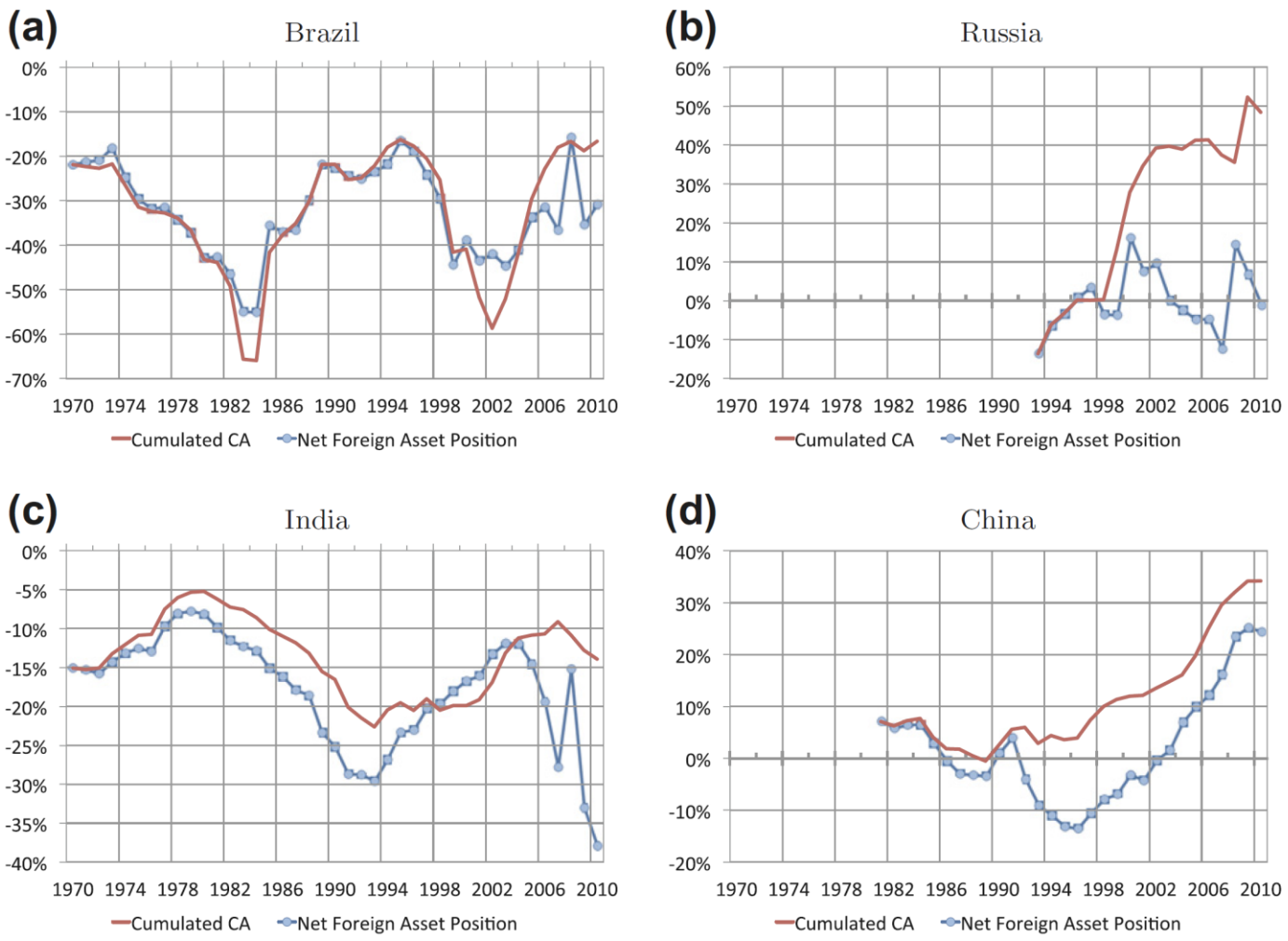


Figure 10.7 Cumulated Current Account and Net Foreign Asset Position: Brazil, Russia, India, and China, 1970–2010. Percent of GDP. *Source:* Lane and Milesi-Ferretti (2007a) updated to 2010.

Source: Gourinchas, Rey (2014) "External Adjustment, Global Imbalances, Valuation Effects"

$\Delta NIIP = CA + valuation\ changes$

$$NIIP = A - L$$

– where A – assets, L – liabilities

- A stylized example where the U.S. is home country
  - The U.S. owns 25 stocks of Eni S.p.A. worth 2 EUR each
  - The U.S. owes 80 bonds worth $1 each
  - Exchange rate: E = 2$/EUR (***notation:* $2 for 1 EUR**)
  - $NIIP = A - L = 2\$/EUR * 2EUR * 25 - \$80 = \$20$
- How does the NIIP change when
  - $ appreciates to E = 1 $/EUR?
  - Eni's stocks jump to 4 EUR?

Recent Data

Valuation Effect is Negative

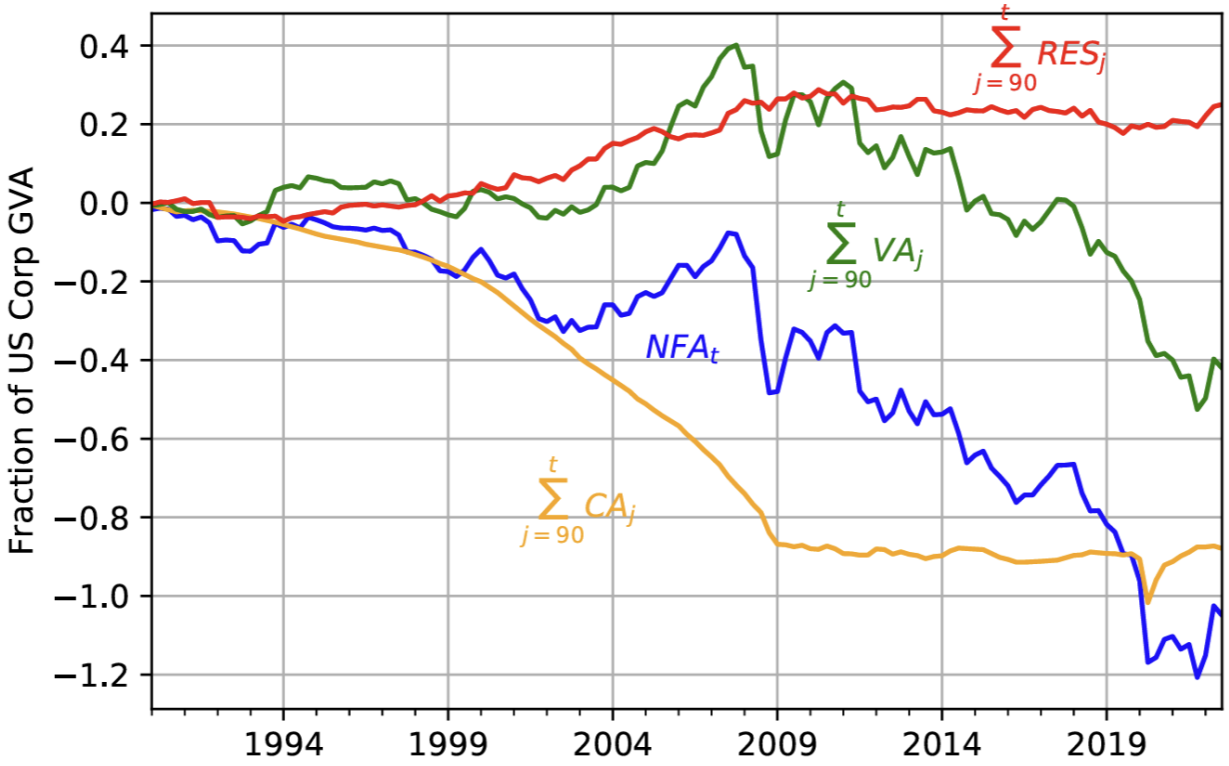


Figure 2: Decomposition of Changes in U.S. NFA over U.S. Corporate Value Added

Source: Atkeson, Heathcote, Perri (2024)

Recent Data

Valuation Effect due to Risky assets turned negative

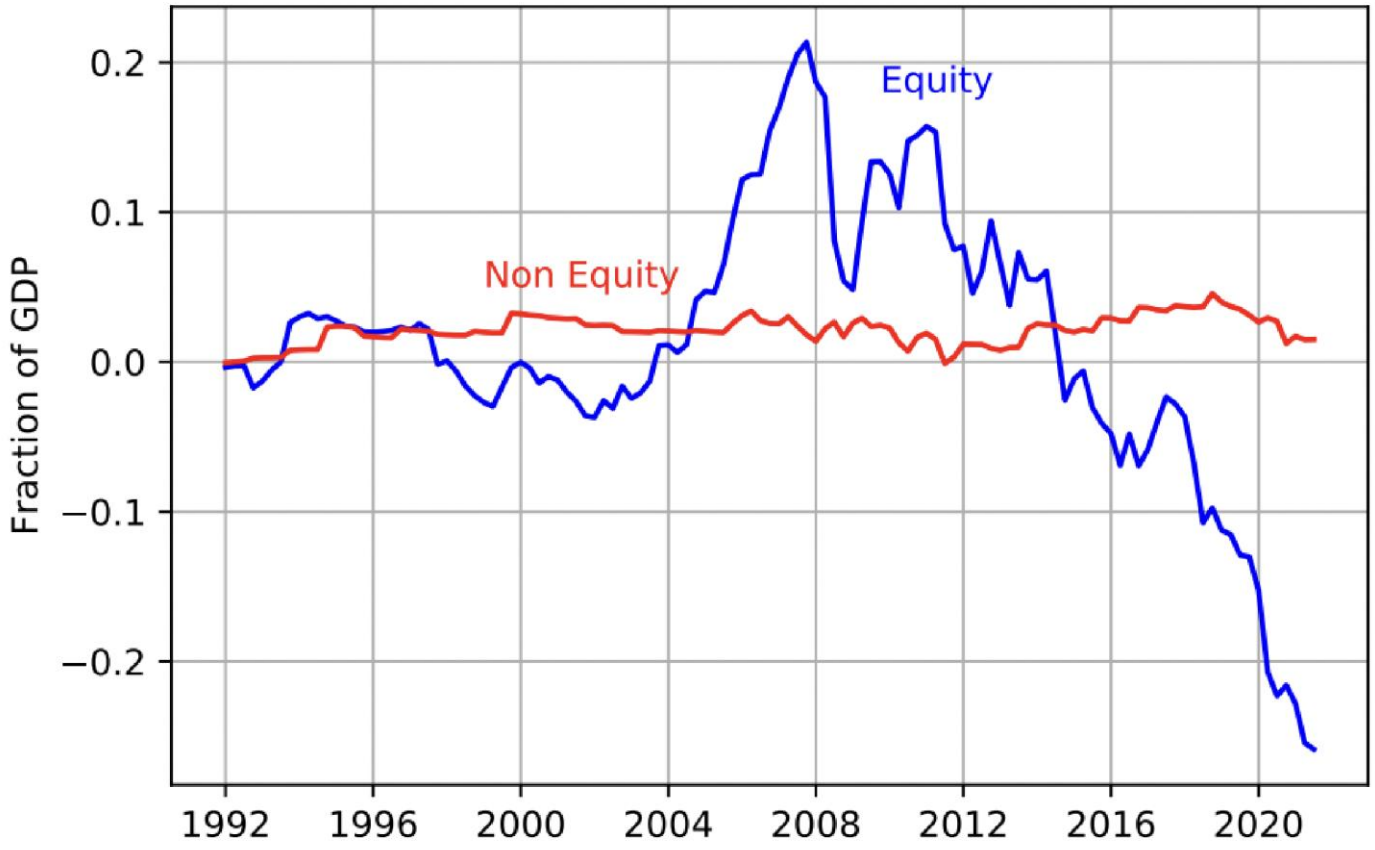
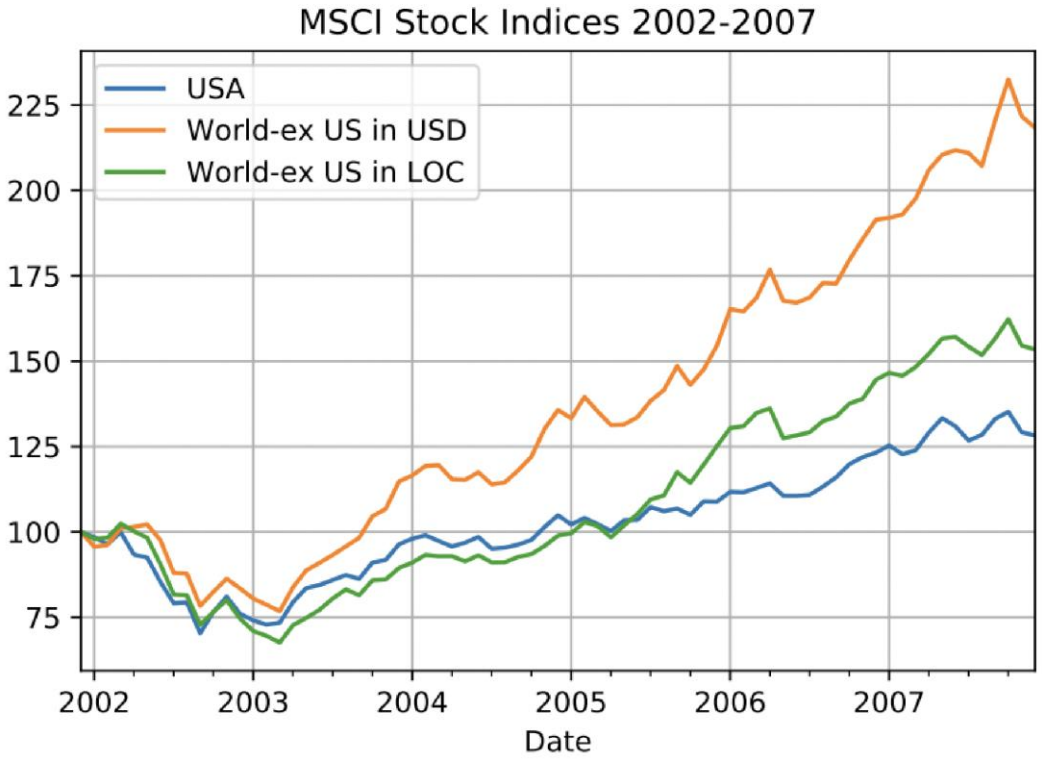
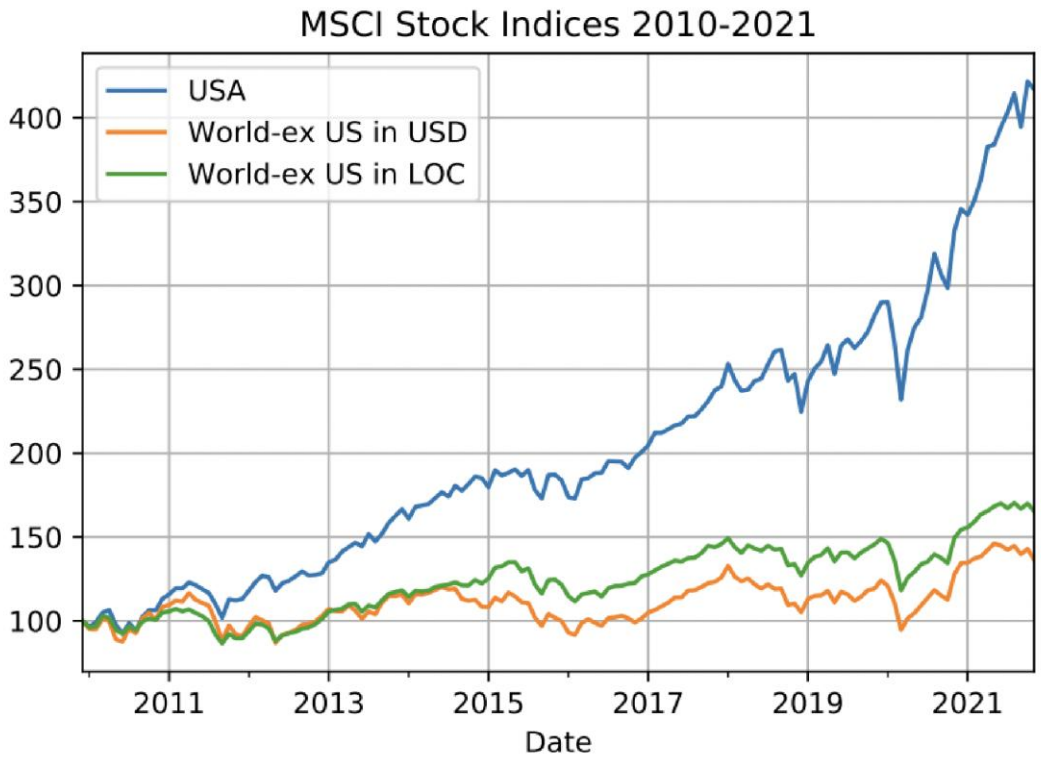


Figure 4: Cumulated Valuation Changes in Equity and Non-equity Positions

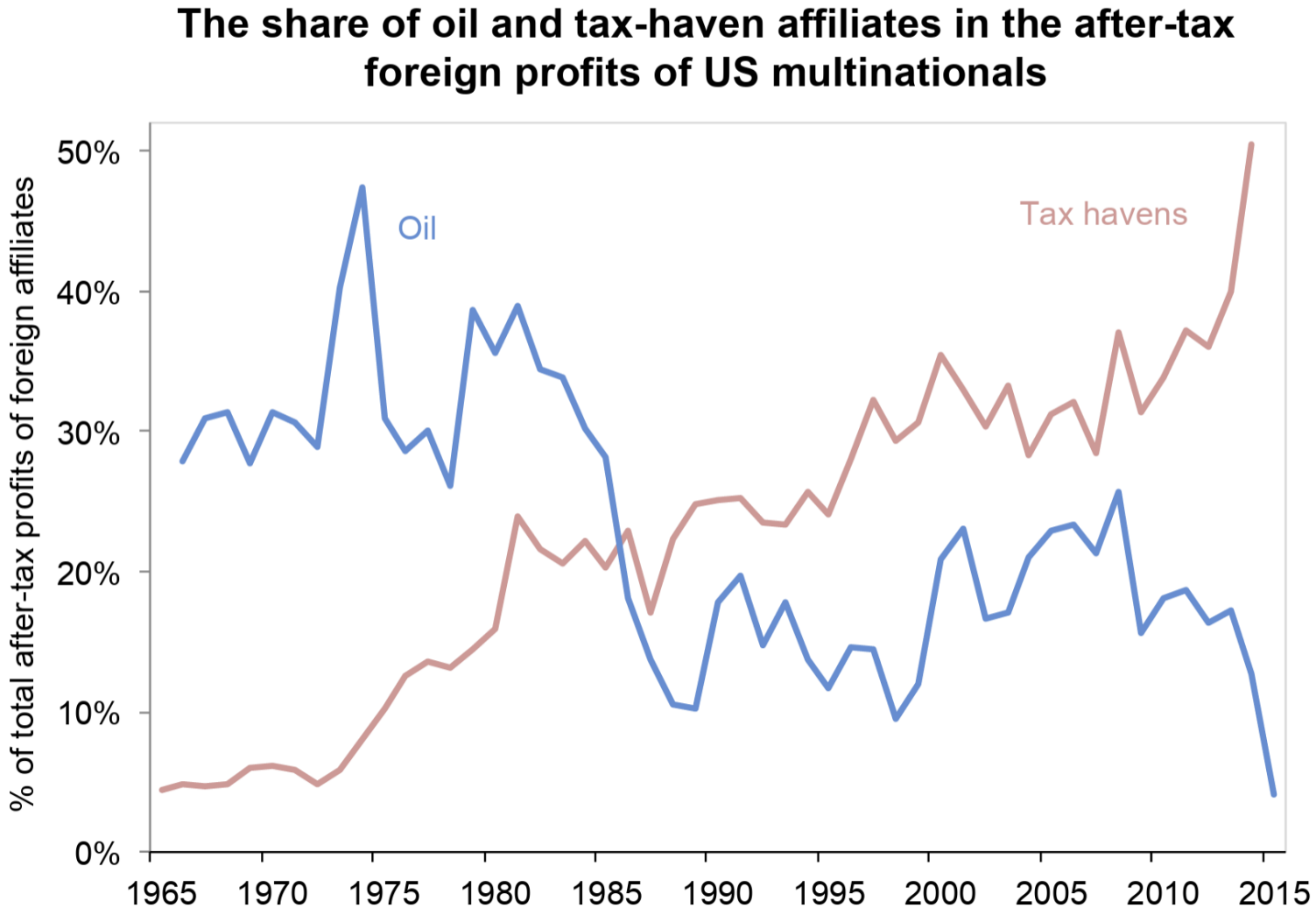
US Stocks underperform before and outperform after GFC





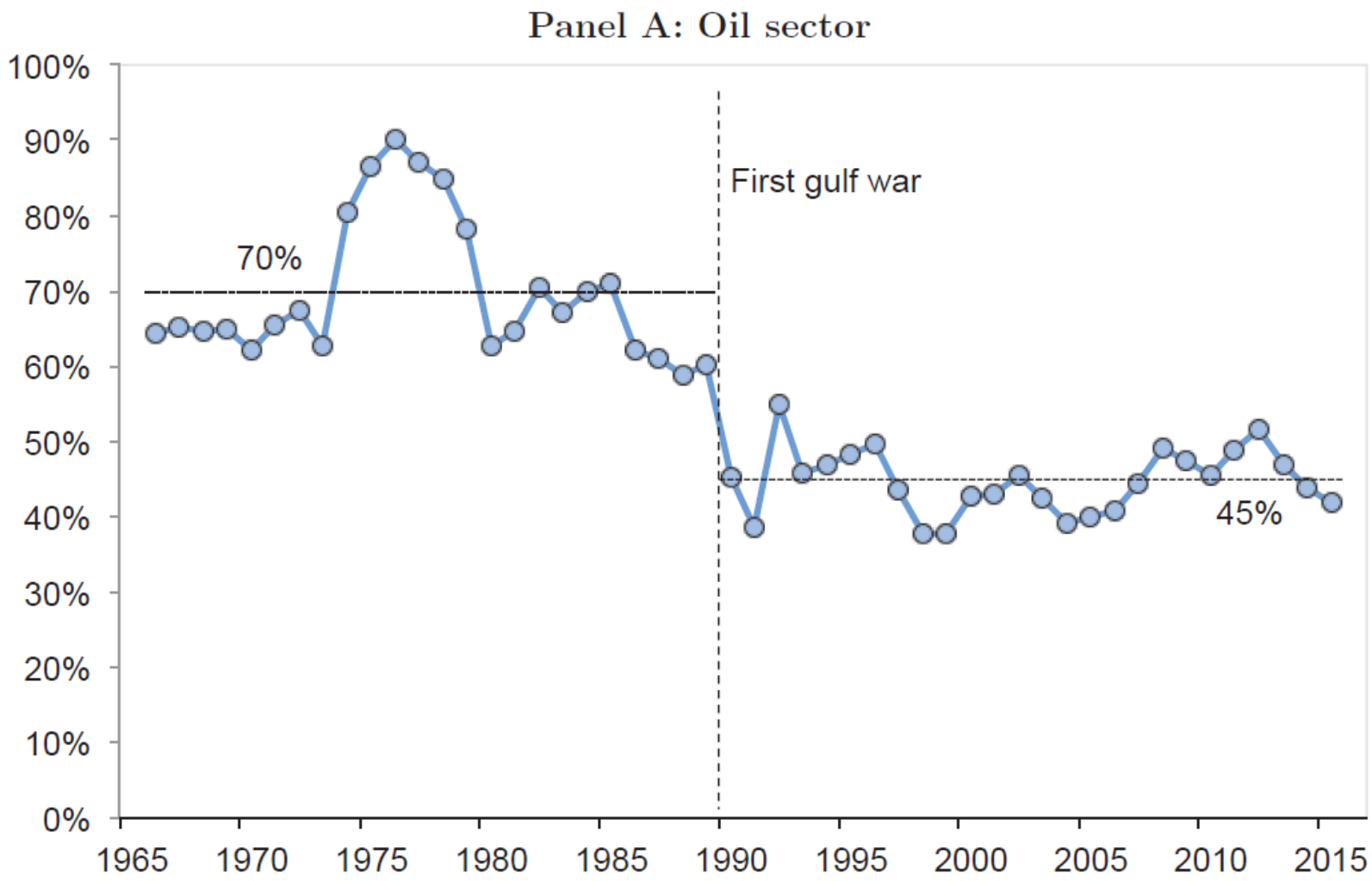
Exorbitant Tax Privilege

US multinationals make over 50% of their foreign profits in oil & tax haven countries



Source: Wright and Zucman (2018)

Tax rates paid by U.S. multinationals on foreign profits

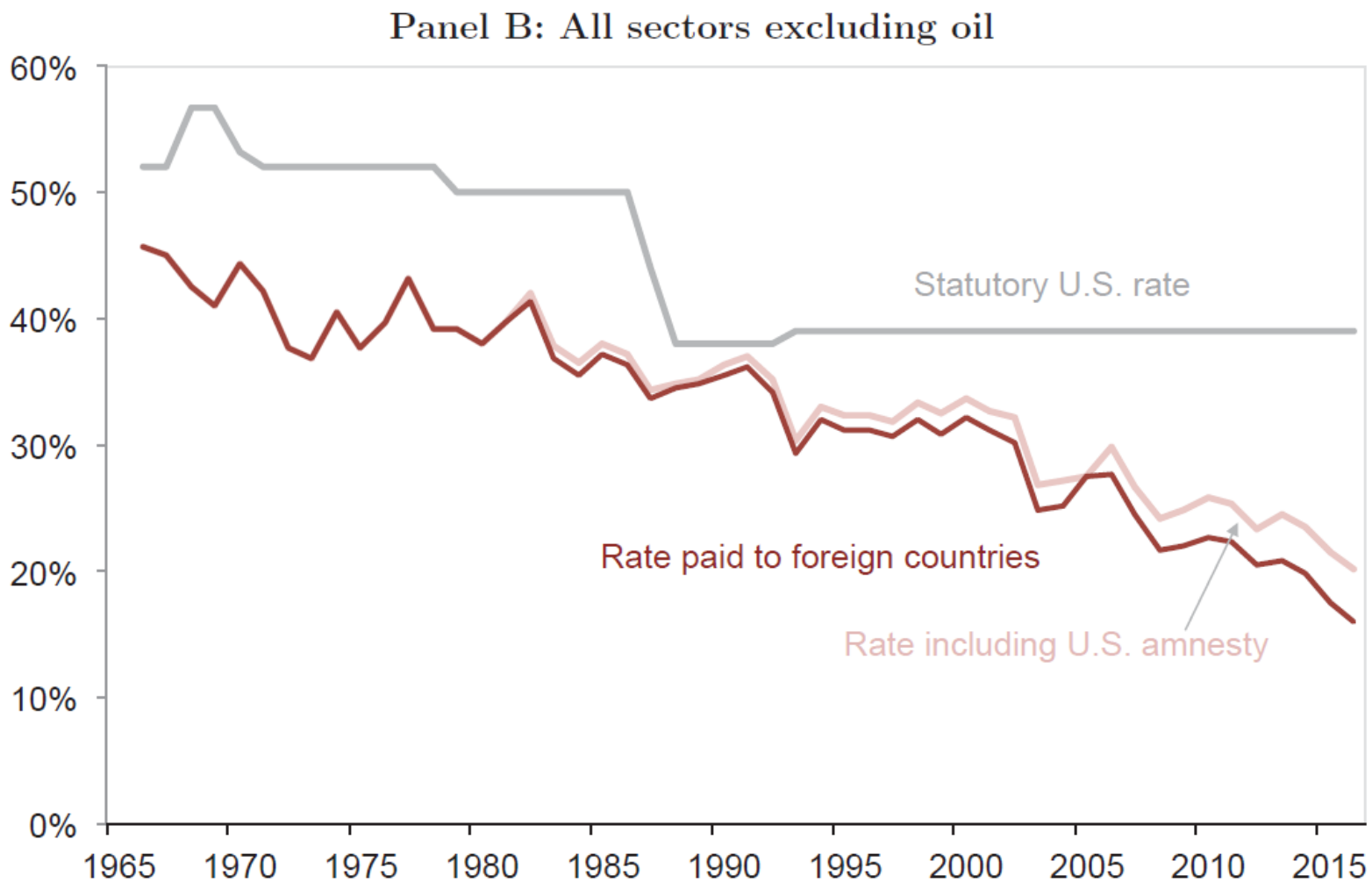


Why did the taxes paid by US oil multinationals abroad fall after 1990?

- Return on military protection granted by the US to oil-producing states
- Effect of tax incentives in response to the oil slump of the mid-1980s
- Tax competition among oil-producing countries

Source: Wright and Zucman (2018)

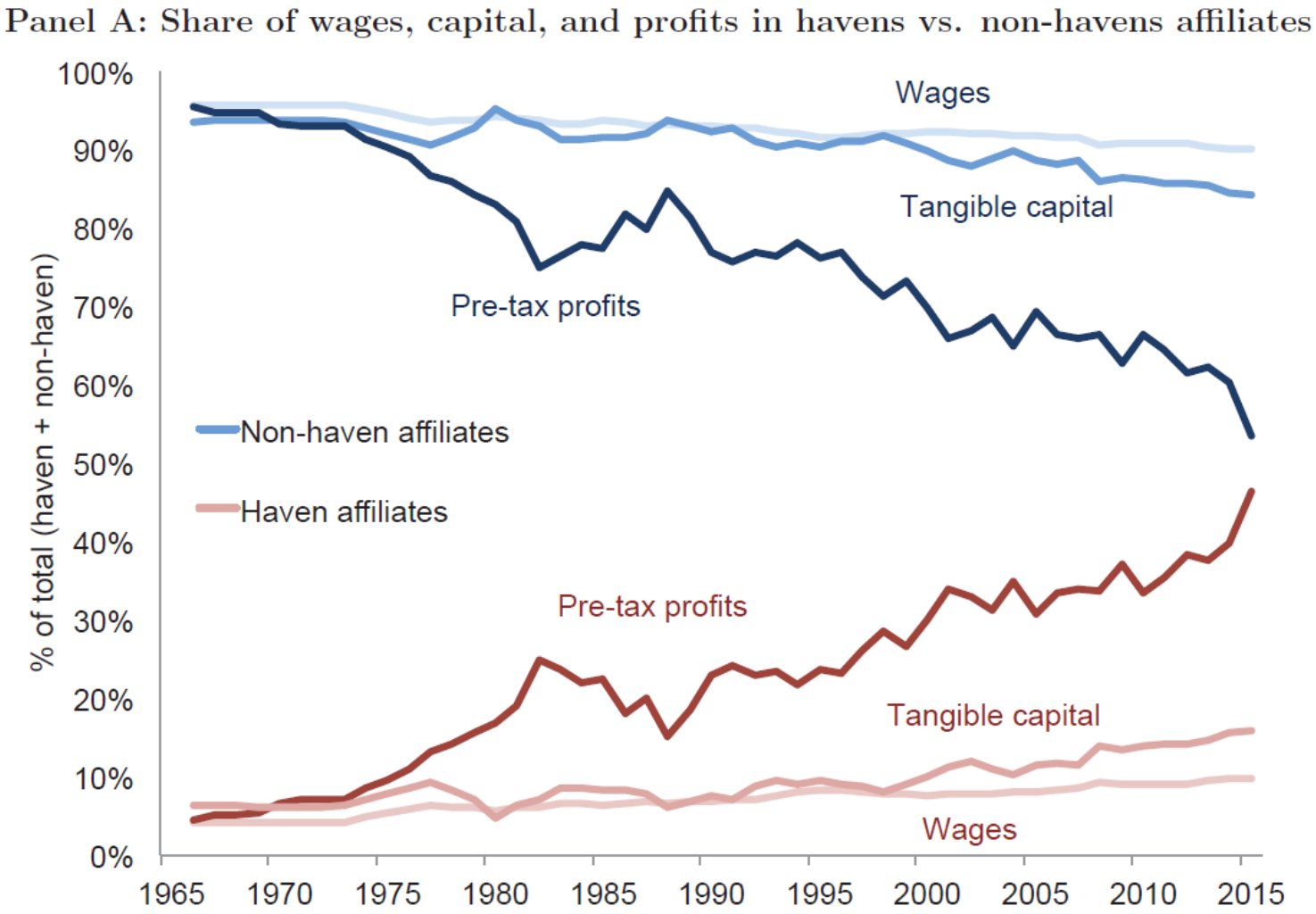
Tax rates paid by U.S. multinationals on foreign profits



- U.S. foreign firms paid smaller foreign taxes on profits than the statutory U.S. tax rates
- Liable for the U.S. taxes only when profits are repatriated to the U.S. Hence, incentives to keep profits abroad.
- **The 2017 Tax Cuts and Jobs Act** lowered the tax rate for repatriated profits from 35% to 8%

Source: Wright and Zucman (2018)

What explains a fall in the foreign tax rate paid by non-oil affiliates?

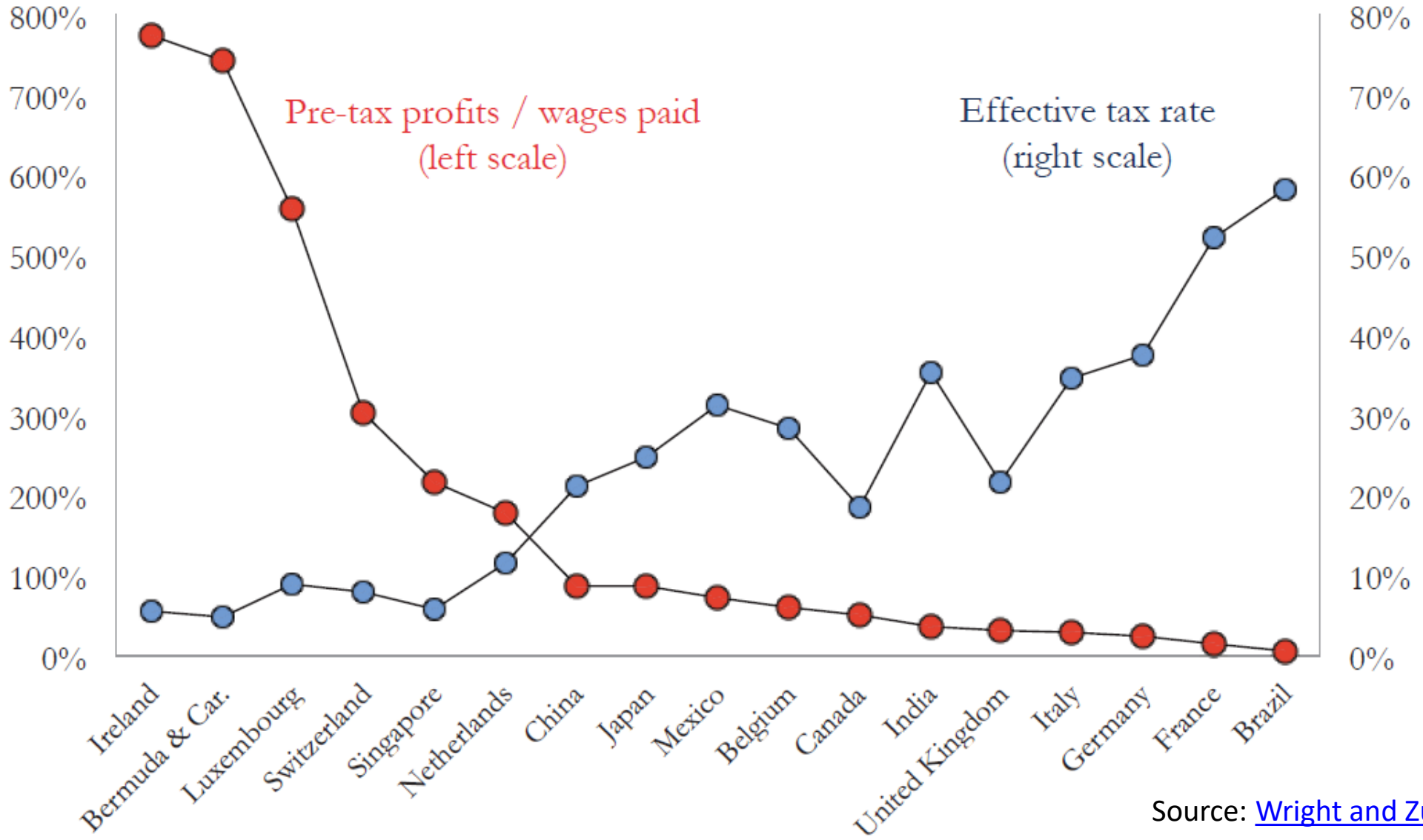


1. Decline in the tax rate in non-haven countries

2. **Profit shifting** to havens with low tax rates
   - *Transfer pricing*: non-market prices within multinationals
   - *Intra-group borrowing and interest payments*
   - *Location of intangibles* (trademarks, patents, logos, etc.)

Source: Wright and Zucman (2018)

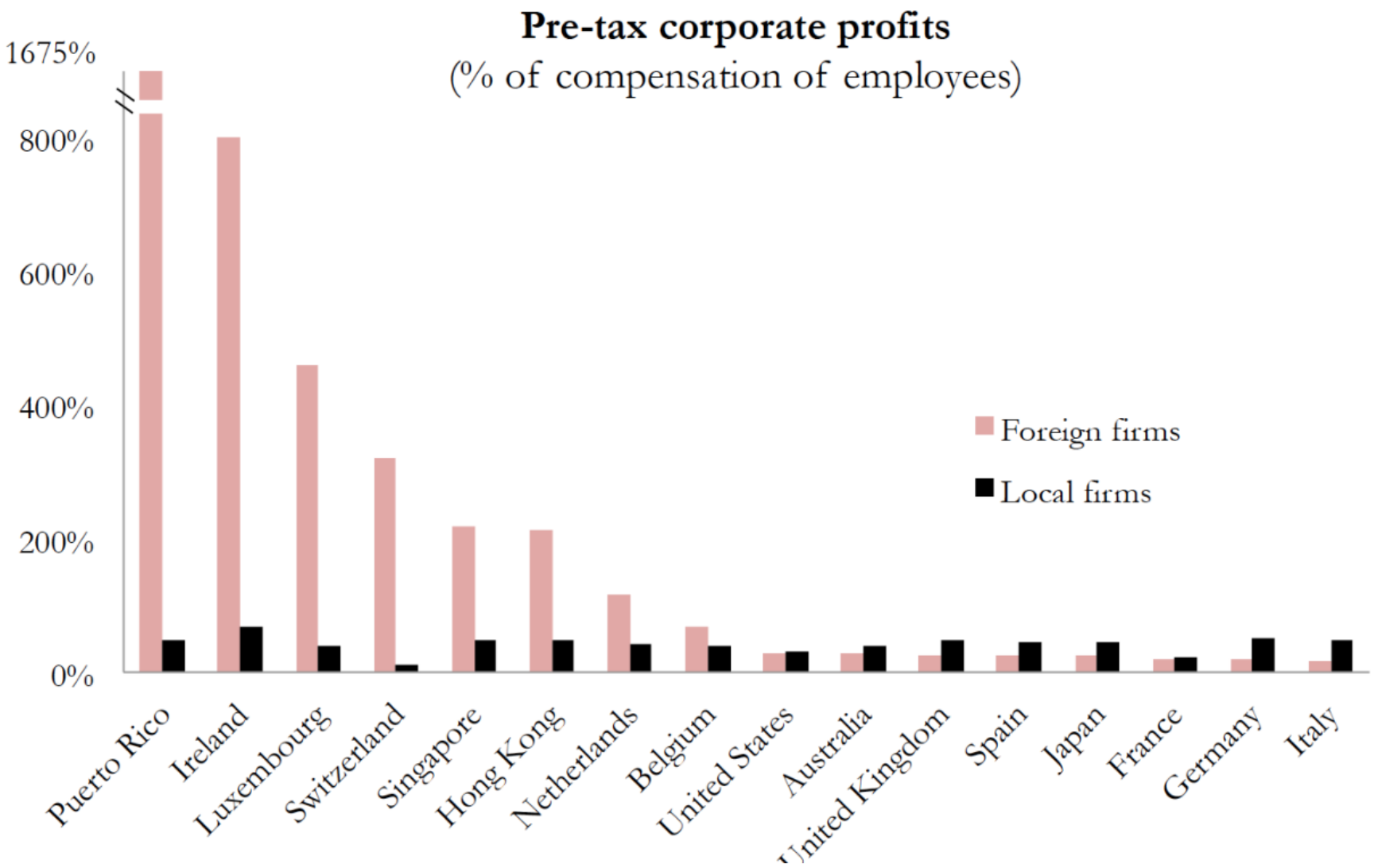
Profit Shifting: pre-tax profits vs. effective tax rates



Non-oil US firms are most profitable (pre-tax) where corporate taxes are lower

Source: [Wright and Zucman (2018)]

Profit Shifting: pre-tax profits: local vs. foreign firms



Foreign firms are consistently less profitable than local firms in non-haven countries but significantly more profitable in tax havens.

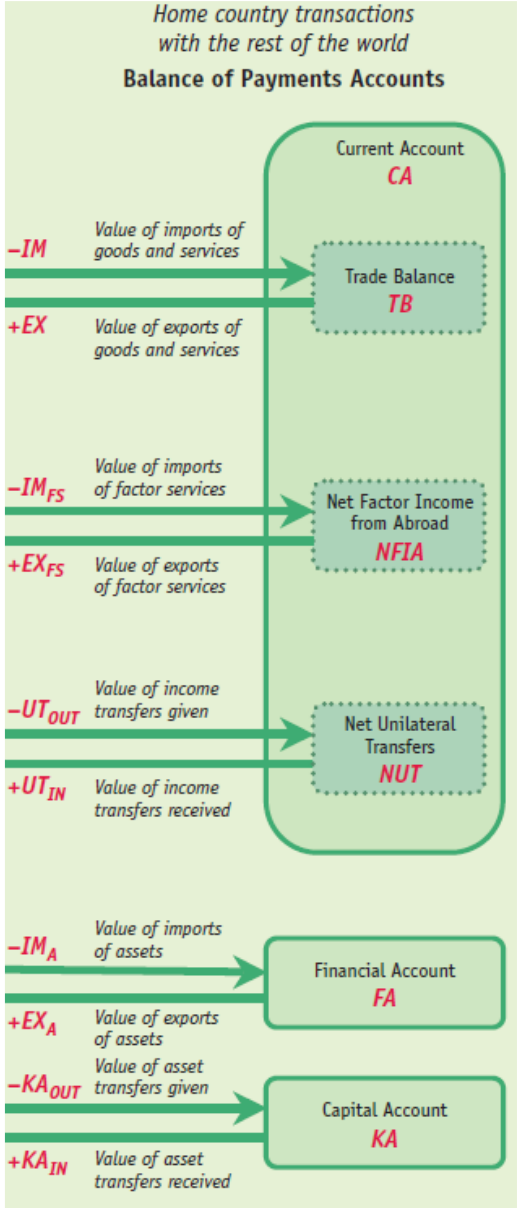
Source: Torslov, Wier and Zucman (2018)

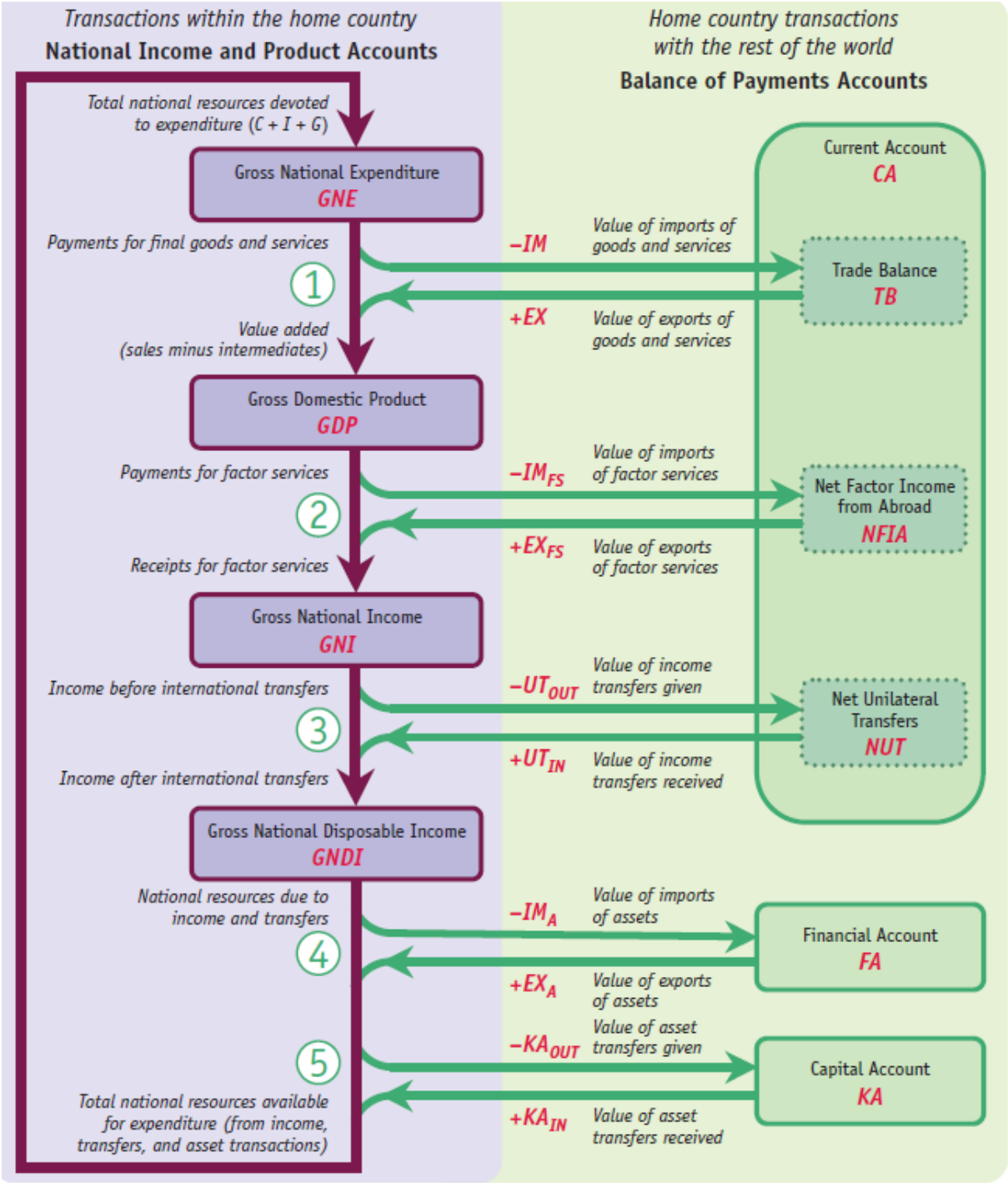
The Balance of Payments Accounts

The Balance of Payments Accounts

- A country's balance of payments accounts keep track of both its payments to and its receipts from foreigners.

- An accounting convention
  - Any payment to foreigners enters with a "-"
  - Any payment from foreigners enters with a "+"




Transactions within the home country
National Income and Product Accounts
Home country transactions with the rest of the world
Balance of Payments Accounts
Total national resources devoted to expenditure (C + I + G)
Gross National Expenditure
GNE
Payments for final goods and services
1
Value added (sales minus intermediates)
Gross Domestic Product
GDP
Payments for factor services
2
Receipts for factor services
Gross National Income
GNI
Income before international transfers
3
Income after international transfers
Gross National Disposable Income
GNDI
National resources due to income and transfers
4
5
Total national resources available for expenditure (from income, transfers, and asset transactions)
Current Account
CA
−IM
Value of imports of goods and services
+EX
Value of exports of goods and services
Trade Balance
TB
−IMFS
Value of imports of factor services
+EXFS
Value of exports of factor services
Net Factor Income from Abroad
NFIA
−UTOUT
Value of income transfers given
+UTIN
Value of income transfers received
Net Unilateral Transfers
NUT
−IMA
Value of imports of assets
+EXA
Value of exports of assets
Financial Account
FA
−KAOUT
Value of asset transfers given
+KAIN
Value of asset transfers received
Capital Account
KA

The Balance of Payments Identity

$$FA + CA + KA = 0$$

- Any international transaction automatically gives rise to two offsetting entries in the balance of payments resulting in an identity

- Sometimes this identity looks like

$$FA = CA + KA$$

  – If the financial account balance is defined with the opposite sign

Example 1

- ITA Airways buys a new Boeing 737 for 90 million €

Example 1

- ITA Airways buys a new Boeing 737 for 90 million €
- A negative entry in the *current account* (import of a good)
- A positive entry in the *financial account*. Why?
  - Boeing receives Euros and needs to do something with them (put them in a bank account, buy European stocks, etc.). No matter what it does, it ends up acquiring a Euro-denominated asset worth 90 million €

- What if ITA Airways pays in US dollars?
  - In this case, ITA Airways needs to get dollars first, and it can do so by selling Euros or Euro-denominated assets.

Example 2

- An Italian tourist Erika visits Sydney and buys a ticket to the Sydney opera for 70€ using her Banca Monte Dei Paschi Di Siena credit card

Example 2

- An Italian tourist Erika, visits Sydney and buys a ticket to the Sydney Opera for 70€ using her Banca Monte Dei Paschi Di Siena credit card

- A negative entry in the *current account* (import of a service)
- A positive entry in the *financial account*. Why?
  - The Sydney Opera House has acquired a Euro-denominated asset: a claim on a payment from an Italian bank

Example 3

- Li Yonghong, a Chinese businessman, buys AC Milan for 740 million €

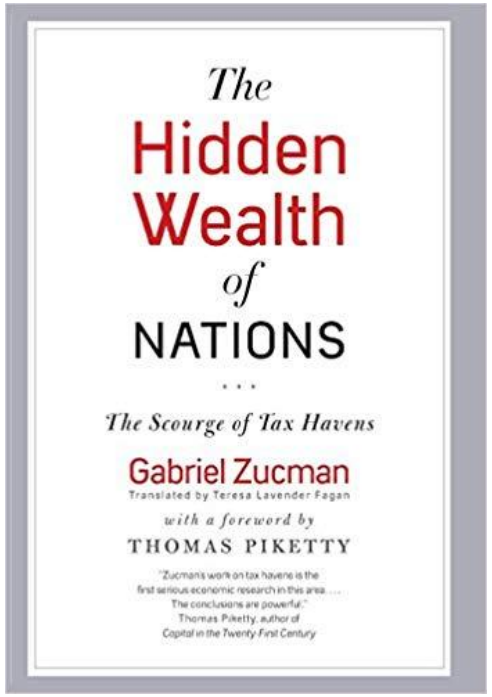
Example 3

- Li Yonghong, a Chinese businessman, buys AC Milan for 740 million €
- A positive entry in the *financial account*
  - A foreigner has acquired an Italian asset.
- A negative entry in the *financial account*
  - in order to get the Euros for the purchase, Li Yonghong (or someone else in China, a bank, for instance) needed to sell some Chinese assets to Europeans.

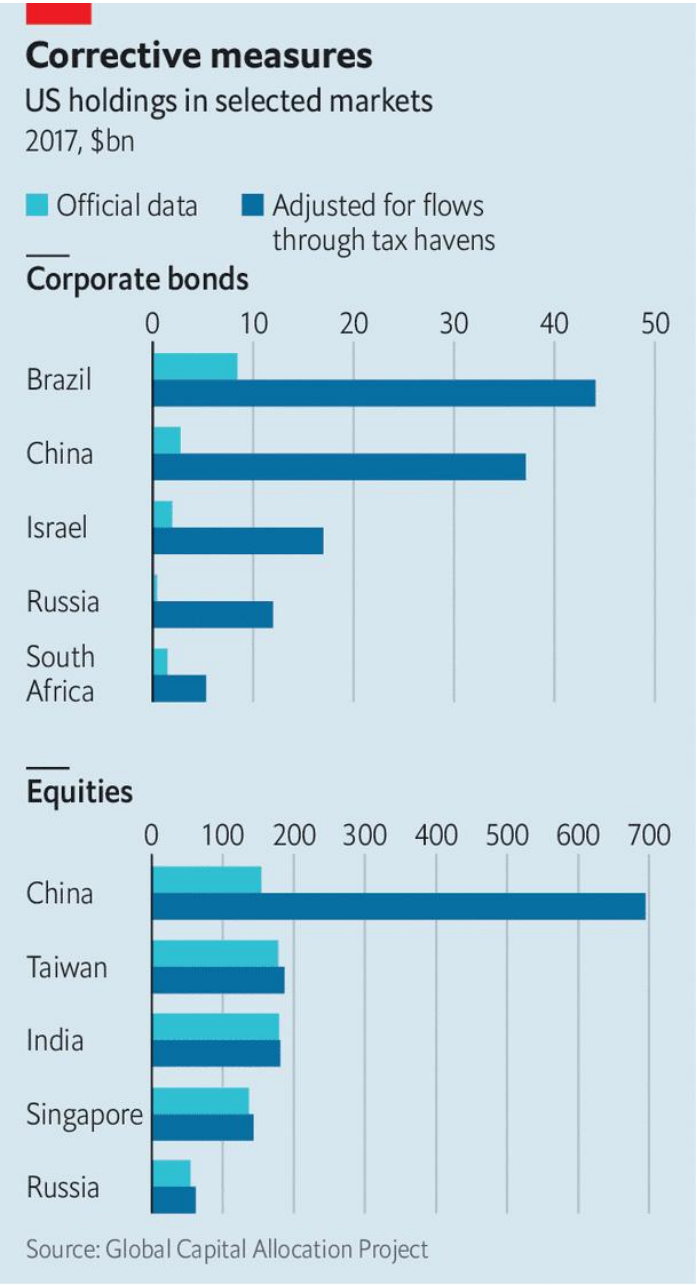
Errors, Omissions and Tax Evasion

- The double accounting principle should ensure that the balance of payments always sums to 0
- In practice, this is not exactly true: errors and non-reported transactions

- It turns out that the world $NIIP$ is negative
  - Using this observation, Gabriel Zucman estimated that 8% of the world's wealth is hidden in offshore accounts
  - This partly explains why the U.S. has a large and negative NIIP



- A huge rise in cross-border investment flowing through tax heavens in the last two decades.
- This distorts official figures. For example,
  - Despite restrictions by Chinese authorities, Americans own a lot of assets in China
- Source: [The Economist]



Next Lecture

- KOM, Chapter 14

Exchange Rates and the Foreign Exchange Market: An Asset Approach

Over the seven years between mid-2007 and mid-2014, the U.S. dollar's price against a basket of major foreign currencies remained roughly constant (despite a temporary rise in the spring of 2009). In the year and a half between mid-2014 and the start of 2016, however, the dollar's value suddenly rose by a whopping 25 percent. What changes in the U.S. and world economy could possibly have driven such a dramatic change in the foreign exchange market? In this chapter we will begin our study of the causes and effects of exchange rate changes.

The price of one currency in terms of another is called an **exchange rate.** At 4 P.M. London time on January 19, 2017 you would have needed 1.0612 dollars to buy one unit of the European currency, the euro, so the dollar's exchange rate against the euro was $1.0612 per euro. Because of their strong influence on the current account and other macroeconomic variables, exchange rates are among the most important prices in an open economy.

Because an exchange rate, the price of one country's money in terms of another's, is also an asset price, the principles governing the behavior of other asset prices also govern the behavior of exchange rates. As you will recall from Chapter 13, the defining characteristic of an asset is that it is a form of wealth, a way of transferring purchasing power from the present into the future. The price an asset commands today is therefore directly related to the purchasing power over goods and services that buyers expect it to yield in the future. Similarly, *today's* dollar/euro exchange rate is closely tied to people's expectations about the *future* level of that rate. Just as the price of Google stock rises immediately upon favorable news about Google's future prospects, so do exchange rates respond immediately to any news concerning future currency values.

Our general goals in this chapter are to understand the role of exchange rates in international trade and to understand how exchange rates are determined. To begin, we first learn how exchange rates allow us to compare the prices of different countries' goods and services. Next, we describe the international asset market in which currencies are traded and show how equilibrium exchange rates are

378